INDUSTRIAL DEFENSE CELL

AI Smart Safety Shield

Hackathon Project Submission Proposal & Technical Blueprints

`Domain: codevortex.in | Global World Hackathon Standard`

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Industrial manufacturing spaces contain heavy machinery capable of causing catastrophic, life-altering trauma or fatalities in milliseconds. Standard mechanical cages, optical light curtains, and safety mats fail because they are high-cost, inflexible, and easily bypassed by plant operators seeking to optimize their workflow times.

The **AI Smart Safety Shield** is a modern, low-cost computer vision lockout framework that establishes dynamic safety envelopes around active hazards. Fusing edge-processed neural networks, cloud state gateways, and low-latency microcontrollers, it detects human operator intrusions and halts physical machinery in milliseconds.

> **Core Impact Metric:** Prevents plant accidents through real-time localized tracking while creating persistent database trails, ensuring 100% industrial safety compliance audits.

2. SYSTEM ARCHITECTURE

The architecture combines local low-latency components with cloud identity and synchronization stacks:

```
+-------------------------+ HTTP / JWT Auth +-----------------------+ | Web UI Portal |
<=======================> | Express Backend Gateway| | (dashboard.html) | | (Render Hosted) | +----------
--------------+ +-----------------------+ ^ || | MJPEG Feed || MongoDB Driver v \/ +--------------------
-----+ Low-Latency Serial +-----------------------+ | AI Edge Neural Loop | <=======================> |
MongoDB Cloud Atlas | | (YOLOv8 + OpenCV) | (UART Commands) | (Persistent History) | +--------------------
-----+ +-----------------------+ || || Contactor Control \/ +-------------------------+ | ESP32 Lockout
Relay | ===> [ Machinery Power Supply Contactor Cut ] +-------------------------+
```

A. Edge AI Service: Runs locally on an edge computer connected directly to IP/CCTV cameras. It intercepts frames, feeds them to YOLOv8 posture detectors, and outputs serial signals.

B. Cloud Gateway Backend: Node.js Express API that registers devices, tracks history events, verifies tokens, and secures camera credentials via AES-256-CBC encryption.

C. Hardware Intercept Unit: An ESP32 microcontroller acting as a physical interlock, pulling machine contactor coils open or shut upon instructions.

3. STEP-BY-STEP FUNCTIONAL WORKFLOWS

Workflow A: Setup & Secure Enrollment

1. An administrator adds a camera configuration using the web dashboard.
2. The frontend checks for private vs. public IP configurations. Private streams are assigned to the Edge Gateway.
3. The Node.js server encrypts the camera access password using AES-256-CBC before saving the record to the Mongo cluster.
4. The dashboard fetches the camera record and initializes the AI video element.

Workflow B: Vision Processing & Proximity Decisions

The Edge AI loop handles frame acquisition, skipping, and zone math:

```
// 1. Calculate Human Bounding Box Center Center C_hx = (X1 + X2) / 2 C_hy = (Y1 + Y2) / 2 // 2. Compute Euclidean Distance to Mapped Danger Core Distance = sqrt( (C_hx - Machine_Cx)^2 + (C_hy - Machine_Cy)^2 ) // 3. Evaluate safety boundaries If Distance <= DANGER_THRESHOLD: State = DANGER ===> Machine STOP signal sent Else if Distance <= WARNING_THRESHOLD: State = WARNING ===> Warning tone sent Else: State = SAFE ===> Normal operations
```

Workflow C: ESP32 Hardware Contactor Lockout

When the AI engine determines a `DANGER` condition:

1. It writes the bytes `"DANGER\n"` to the active COM port.
2. The ESP32 microcontroller reads the buffer, sets the relay GPIO pin `HIGH`, which immediately opens the magnetic contactor circuit, cutting power to the machinery.
3. When the operator clears the danger zone, the command `"SAFE\n"` is sent, allowing the system to resume once the operator presses the physical reset button.

4. ROLE-BASED ACCESS MATRIX

To prevent plant operators from overriding safety lockouts or deleting logs, the platform implements a strict role-based access matrix:

System Capability	Viewer	Plant Operator	Administrator
View Solution Landing Page	ALLOWED	ALLOWED	ALLOWED
Access Live Stream Monitor	DENIED	ALLOWED	ALLOWED
Add / Register IP Nodes	DENIED	DENIED	ALLOWED
Trigger Emergency Manual Trips	ALLOWED	ALLOWED	ALLOWED
Clear Intrusion Log History	DENIED	DENIED	ALLOWED

5. CORE IMPLEMENTATIONS

AI Server Telemetry Loop (Python snippet)

```
# danger_zone_detection.py - Core Decision Block if self.safety_state == "DANGER": if self._last_safety_state in ["SAFE", "WARNING"]: trigger_email = True if trigger_email: # Dispatch warning payload with base64 snapshot to Cloud Gateway payload = { "danger": True, "confidence": int(ai_confidence), "cameraStreamUrl": str(self.source), "image": img_b64, "recipient_email": self.recipient_email } dispatch_cloud_alert(payload)
```

ESP32 Controller Firmware (C++ Code)

```
// ESP32 Contactor Controller const int RELAY_PIN = 5; void setup() { Serial.begin(115200); pinMode(RELAY_PIN, OUTPUT); digitalWrite(RELAY_PIN, LOW); // Default Safe On Start } void loop() { if (Serial.available() > 0) { String command = Serial.readStringUntil('\n'); command.trim(); if (command == "DANGER") { digitalWrite(RELAY_PIN, HIGH); // Open contactor (Cut Power) } else if (command == "SAFE") { digitalWrite(RELAY_PIN, LOW); // Close contactor (Restore Power) } } }
```

6. COMMERCIALIZATION & FUTURE EXPANSION

The AI Smart Safety Shield represents a highly scalable safety technology. Future updates planned for scale include:

- **Dynamic Mesh Contours:** Allowing operators to draw safety boundaries using an interactive map, replacing rigid static rectangles.
- **Thermal Signature Integration:** Using dual-spectrum thermal cameras to detect human warmth patterns, bypassing steam, smoke, or heavy dust barriers.
- **Sensor Fusion:** Integrating ultrasonic sensor ranges with visual neural nodes to create redundancy checks.